PROOF OF THE PLANAR PÓLYA–SZEGŐ BUCKLING CONJECTURE

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1. Introduction

For an open set $\Omega \subset \mathbb{R}^2$ of finite positive area, define

$$\Lambda_1(\Omega) = \inf_{0 \neq v \in H_0^2(\Omega)} \frac{\int_\Omega (\Delta v)^2 \, \mathrm{d}x}{\int_\Omega |\nabla v|^2 \, \mathrm{d}x}, \qquad \lambda_1^{\mathrm{D}}(\Omega) = \inf_{0 \neq v \in H_0^1(\Omega)} \frac{\int_\Omega |\nabla v|^2 \, \mathrm{d}x}{\int_\Omega v^2 \, \mathrm{d}x}.$$

Here $H_0^m(\Omega)$ denotes the closure of $C_c^\infty(\Omega)$ in $H^m(\Omega)$. This definition is used throughout, including when $\partial\Omega$ is irregular. For a smooth bounded domain, the first quantity is the least eigenvalue of the clamped buckling problem

$$\Delta^2 v + \Lambda \Delta v = 0 \quad \text{in } \Omega, \qquad v = \partial_\nu v = 0 \quad \text{on } \partial\Omega,$$

where ν is the outward unit normal. In the planar plate model, Λ_1 represents the critical load for uniform lateral compression.

The isoperimetric question for this eigenvalue belongs to the classical program of comparing membranes and plates of prescribed area. The Faber–Krahn theorem [8, 11] identifies the disk as the minimizer of λ_1^{D}. Pólya and Szegő [16] proposed the corresponding assertion for the first buckling load. Let J_ν be the Bessel function of the first kind, and let $j_{\nu,n}$ denote its nth positive zero. On a disk of radius R,

$$\lambda_1^{\mathrm{D}}(B_R) = j_{0,1}^2/R^2, \qquad \Lambda_1(B_R) = j_{1,1}^2/R^2.$$

The planar Pólya–Szegő conjecture takes the following form.

Conjecture 1.1. *For every smooth bounded planar domain Ω,*

$$|\Omega| \Lambda_1(\Omega) \geq \pi j_{1,1}^2,$$

with equality if and only if Ω is a disk.

Szegő's early symmetrization argument [19, 20] established the buckling inequality under an additional hypothesis that a first eigenfunction does not change sign. The correction in [20] addresses the admissibility of the rearranged function. The sign hypothesis cannot be imposed on general buckling eigenfunctions; see the discussion in [3] and the analysis of annuli in [5]. Thus the usual positivity-based proof for the Dirichlet Laplacian does not directly settle Conjecture 1.1.

A second line of work compares buckling and membrane spectra. Payne's inequality $\Lambda_1 \geq \lambda_2^{\mathrm{D}}$ [15], with the correction discussed by Friedlander [9], gives a useful but nonsharp area bound. Here λ_j^{D} denotes the jth Dirichlet eigenvalue, counted with multiplicity. In dimension two, the bound of Bramble and Payne [4] is $|\Omega|\Lambda_1 > 2\pi j_{0,1}^2$ for bounded connected domains. Ashbaugh and Laugesen [3] extended this estimate to higher dimensions. More recently, Levitin and Polterovich [13] established Payne's shifted comparison for every odd buckling index. These index comparisons are distinct from the comparison with λ_1^{D} proved below.

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Shape optimization provides another approach. Ashbaugh and Bucur [2] proved existence results in the simply connected planar class and for a relaxed problem. Stollenwerk [17] subsequently proved existence among finite-measure open sets in arbitrary dimension. The Weinberger–Willms argument, presented in [2], identifies a smooth simply connected optimizer as a disk; related optimality conditions were developed by Stollenwerk and Wagner [18]. Lamboley and Nahon [12] proved boundedness and partial boundary regularity of planar optimizers, and weakened the regularity assumptions in this characterization. Those results leave hypotheses on the optimizer; the argument here does not require an optimal-domain regularity theorem.

Our main result is a sharp comparison with the first Dirichlet eigenvalue.

Theorem 1.2. *For every open set $\Omega \subset \mathbb{R}^2$ with $0 < |\Omega| < \infty$,*

$$\Lambda_1(\Omega) \geq \frac{j_{1,1}^2}{j_{0,1}^2} \lambda_1^{\mathrm{D}}(\Omega). \tag{1}$$

The constant is optimal.

Combining Theorem 1.2 with the Faber–Krahn inequality resolves Conjecture 1.1, without assumptions on topology or boundary regularity.

Corollary 1.3. *For every open set $\Omega \subset \mathbb{R}^2$ of finite positive area,*

$$|\Omega| \Lambda_1(\Omega) \geq \pi j_{1,1}^2. \tag{2}$$

Equality holds if and only if Ω is an open disk as a set.

The proof of (1) is variational. We construct a one-variable function f for which

$$F = |\nabla U| f\big(U/|\nabla U|\big)$$

converts a compactly supported buckling test function into a Dirichlet test function. The value at $\nabla U = 0$ is specified by continuity. After adding two divergence terms whose integrals vanish, the desired functional inequality reduces to positivity of a symmetric matrix. Its kernel is determined by the radial disk eigenfunctions. The remaining 2×2 block is controlled by convexity, two Wronskian comparisons, and a residue expansion for an auxiliary Bessel function. The construction is compatible with changes of sign of U. In particular, it does not use positivity of a buckling eigenfunction.

Section 2 defines the profiles and proves the matrix identities. Sections 3–6 establish the one-variable inequalities. Section 7 treats the weak interpretation of the divergence terms and proves Theorem 1.2. Section 8 proves the equality statement. The appendices collect the fixed Bessel estimates and the form-domain facts used in the proof. All functions may be taken real-valued: the quotient of a complex function is a weighted average of the quotients of its nonzero real and imaginary parts.

Lean formalization. A Lean 4 formalization of the main results is available at `https://github.com/QuanyuTang/polya-szego-planar-unconditional-palomar`. The formal development proves the sharp buckling–Dirichlet comparison in Theorem 1.2, the area inequality in Corollary 1.3, the exact characterization of equality by an open disk, and the buckling eigenvalue formula for disks. Its variational definitions use the closures of smooth compactly supported functions in the corresponding Sobolev norms, including for irregular open sets.

The formalization has been verified and registered in the Palomar Registry (entry PALOMAR-2026-10-18-000008, version 1). The repository provides the complete Lean sources, pinned dependencies, reproducible build instructions, verification records, and an audit of the axioms used by the final declarations.

The formalization was developed from the authors' original informal proof. It retains the central functional comparison based on Bessel profiles, divergence identities, and matrix positivity, but it is not a line-by-line transcription of the argument in this paper. In particular, the formal development includes separate proofs of the Faber–Krahn inequality, its equality characterization, and the additional rigidity needed for exact equality of open sets. The formal verification establishes the stated mathematical results, rather than verifying every individual step of the informal proof presented here.

Use of artificial intelligence. The mathematical strategy and the original informal proof presented in this paper were developed by the authors with assistance from ChatGPT in exploring technical arguments and preparing the manuscript. Harmonic's Aristotle and OpenAI's Codex constructed the Lean formalization based on the authors' original informal proof. The formal development was subsequently refined and checked using Lean 4; as explained above, it is not a line-by-line formalization of this manuscript. The authors take responsibility for the mathematical statements, the informal proof, and the accuracy of the description of the formalization.

2. The auxiliary profiles and matrix identity

2.1. **The disk profiles.** Put $k = j_{1,1}$, $q = j_{0,1}$, $\alpha = q/k$, $\lambda = \alpha^2$, $c = -J_0(k) > 0$, and, for $0 < r < k$,

$$p = J_1(r), \qquad u = J_0(r) + c, \qquad \phi = J_0(\alpha r),$$
$$A = -p_r/p, \quad C = -1/r, \quad t = u/p, \quad T = t_r = tA - 1.$$

Here $p, u, \phi > 0$. The radial functions u and ϕ are the first buckling and Dirichlet eigenfunctions on B_k, with eigenvalues 1 and λ, respectively. Since $u_r = -p$, the change of variable $t = u/p$ is chosen so that the transform $pf(t)$ sends u to ϕ. Define

$$f(t(r)) = \phi/p, \qquad z = f_t, \qquad g = f - tz, \qquad w = \phi_r/p, \qquad K = \left(\frac{\alpha J_1(q)}{c}\right)^2.$$

A profile written as a function of r means its composition with $t = t(r)$ when necessary; derivatives carry a subscript r or t. Primes in radial calculations denote r derivatives. All terminating decimals below denote exact rational numbers. The Bessel equation, defining series and recurrences are given in [14, Eqs. 10.2.1–10.2.2 and 10.6.1–10.6.3]; the canonical product and interlacing of real zeros are given in [14, Eqs. 10.21.15 and 10.21.2]. Appendix A proves all fixed constant bounds used below, including the identification of their root indices.

2.2. **The change of variable.**

Lemma 2.1. *$T < 0$ on $(0, k)$, and t maps that interval bijectively onto $(0, \infty)$ in reverse order.*

Proof. Let $S = p^2 + up_r$, so $T = -S/p^2$. If $p_r \geq 0$ then $S > 0$. The elementary alternating Bessel bounds $J_0(r) \geq 1 - r^2/4$ and $J_1(r)/r \leq 1/2$ imply $p_r = J_0 - p/r > 0$ on $0 < r \leq \sqrt{5}/2$. If $p_r(r) < 0$, necessarily $r > \sqrt{5}/2$. For $L = p_r/p$, the Bessel ODE gives

$$(\log p)_{rr} = -1 + \frac{1}{r^2} - \frac{L}{r} - L^2 = -1 + \frac{5}{4r^2} - \left(L + \frac{1}{2r}\right)^2 < 0 \quad (r > \sqrt{5}/2).$$

Thus $\log p$ is strictly concave on $[r, k)$ and, since $A(r) > 0$,

$$u(r) = \int_r^k p(s)\, ds < \int_r^\infty p(r) e^{-A(r)(s-r)} ds = \frac{p(r)}{A(r)}.$$

It follows that $S = p^2(1 - tA) > 0$. The endpoint expansions $t \sim 2(1 + c)/r$ at zero and $t \sim (k - r)/2$ at k prove surjectivity. □

The following formulas will be useful:

$$S = p^2 + up_r > 0, \qquad z = \frac{\phi p_r - p\phi_r}{S}, \qquad g = \frac{p\phi + u\phi_r}{S}. \tag{3}$$

The Bessel equations give the identities

$$A_r = A^2 + AC + 1 - C^2, \qquad C_r = C^2, \tag{4}$$

$$f_r = w + Af = Tz, \qquad w_r = (A + C)w - \lambda f, \qquad w = -z - Ag. \tag{5}$$

2.3. **Two auxiliary functions.** Let

$$B(x) = x[J_0(x)^2 + J_1(x)^2] - J_0(x)J_1(x).$$

The recurrence relations give

$$B'(x) = \frac{J_0(x)J_1(x)}{x}.$$

Define

$$G(r) = K + \frac{\alpha r B(\alpha r) - KrB(r)}{p(r)^2}. \tag{6}$$

The choice of K gives

$$B(k) = kc^2, \qquad B(q) = qJ_1(q)^2, \qquad KB(k) = \alpha B(q).$$

This identity will remove the apparent singularity at $r = k$.

Define

$$b = -KA - GC - fw = -KA + \frac{G}{r} - \frac{\phi\phi_r}{p^2}. \tag{7}$$

Lemma 2.2. *The function G satisfies*

$$V := w^2 - Cfw + \lambda f^2 = KA_r + GC^2. \tag{8}$$

One also has

$$G_r = b + 2AG + 2KC, \tag{9}$$

$$b_r + CG_r = -2f_r w. \tag{10}$$

Proof. Expand (6) using the definition of B, multiply by p^2, and use (4). The result is exactly $G = r^2(V - KA_r)$, proving (8). By (5) and $C_r = C^2$,

$$V_r = (2A + C)V - C^2 fw.$$

Also

$$A_{rr} = (2A + C)A_r + AC^2 - 2C^3.$$

Differentiate (8), substitute these two formulas, and cancel $K(2A+C)A_r$. Division by $C^2 > 0$ gives

$$G_r = (2A - C)G - KA + 2KC - fw = b + 2AG + 2KC.$$

Differentiating (7) gives

$$b_r + CG_r = -KA_r - GC^2 - f_r w - fw_r = -V - f_r w - fw_r = -2f_r w,$$

where the last equality uses $f_r w - fw_r = w^2 - Cfw + \lambda f^2 = V$. □

2.4. **Reduction to a two-dimensional quadratic form.** Define the coefficients

$$a = K - g^2, \quad d = 2G - tG_t, \quad L_A = 2b - tb_t - 2zg, \quad L_C = b + G_t, \quad H = -K - z^2 + \lambda f^2 + b_t.$$

Define the symmetric matrix

$$M = \begin{pmatrix} a & d/2 & L_A/2 \\ d/2 & K & L_C/2 \\ L_A/2 & L_C/2 & H \end{pmatrix}.$$

Proposition 2.3. *For all $0 < r < k$,*

$$L_C = -dA - 2KC, \qquad L_A = -2aA - dC, \qquad H = aA^2 + dAC + KC^2.$$

Proof. The first equation follows from (9):

$$dA + 2KC + L_C = 2GA + 2KC + b + (1 - tA)G_t = G_r - TG_t = 0.$$

For the second, multiply its left side by T and use (7), $f - g = tz$, and $w = -z - Ag$:

$$\begin{aligned} T(L_A + 2aA + dC) &= T[2b - 2zg + 2(K - g^2)A + 2GC] - t(b_r + CG_r) \\ &= -2tTzw - t(b_r + CG_r) = 0, \end{aligned}$$

by (10) and $Tz = f_r$. Finally (10) implies $b_t = -2zw - CG_t$. Therefore

$$\begin{aligned} H - (aA^2 + dAC + KC^2) &= \lambda f^2 + w^2 - K(1 + A^2 + C^2) - 2GAC + CG_r \\ &= \lambda f^2 + w^2 - K(1 + A^2 - C^2) + Cb \\ &= V - GC^2 - K(A^2 + AC + 1 - C^2) = 0 \end{aligned}$$

by (9), (7), (4) and (8). □

Proposition 2.4. *If*

$$d^2 + 4Kg^2 \le 4K^2, \tag{11}$$

then M is positive semidefinite and $2K - g^2 - d \ge 0$.

Proof. The kernel identities give the quadratic-form identity

$$(X, Y, P)M(X, Y, P)^T = a(X - AP)^2 + d(X - AP)(Y - CP) + K(Y - CP)^2.$$

The block determinant is $K(K - g^2) - d^2/4$; nonnegativity implies $g^2 \le K$. Moreover

$$2K - g^2 - d \ge 2K - g^2 - 2\sqrt{K(K - g^2)} = \left(\sqrt{K} - \sqrt{K - g^2}\right)^2 \ge 0.$$

□

2.5. **Analytic endpoint data.** Write $\beta = \sqrt{K}$, $a_0 = 1 - \lambda$ and $\delta = k - r$. The Bessel equations give

$$p = c\delta + \frac{c}{2k}\delta^2 + \frac{c(3/k^2 - 1)}{6}\delta^3 + O(\delta^4), \qquad u = \frac{c}{2}\delta^2 + \frac{c}{6k}\delta^3 + O(\delta^4),$$

$$\phi = c\beta\delta + \frac{c\beta}{2k}\delta^2 + \frac{c\beta(2/k^2 - \lambda)}{6}\delta^3 + O(\delta^4).$$

Thus $t = \delta/2 + O(\delta^2)$ and, writing $m_k = 1 - \lambda - k^{-2}$,

$$g = \beta - \frac{\beta m_k}{6}\delta^2 + O(\delta^3), \qquad z = \frac{2\beta m_k}{3}\delta + O(\delta^2).$$

Here $m_k > 0$ follows already from $\lambda < 2/5$ and $k > 19/5$, whose elementary rational justifications are supplied below.

The endpoint values of G and its slope are exact. Put $N(r) = \alpha r B(\alpha r) - K r B(r)$. Differentiating $(sB(s))' = s(J_0(s)^2 + J_1(s)^2)$ gives

$$N(k) = N'(k) = 0, \quad N''(k) = -2Kc^2, \quad N_{rrr}(k) = 2Kc^2/k.$$

Consequently

$$G = \frac{Kp^2 + N}{p^2} = \frac{2K}{3k}\delta + O(\delta^2), \qquad G(k) = 0, \quad G_r(k) = -\frac{2K}{3k}.$$

All these expansions are convergent local analytic expansions after the indicated removable factors have been cancelled. At the radial center, $G|_{r=0} = 2\lambda - K$, $g|_{r=0} = 0$, $z|_{r=0} = (1+c)^{-1}$; these follow from the ordinary Bessel series.

3. Convexity of the transform

Put $r_0 = (1-\lambda)^{-1/2}$. Only the coarse elementary parameter bounds

$$0 < \lambda < \frac{2}{5}, \qquad 0 < c < \frac{41}{100}, \qquad k > \frac{19}{5} \tag{12}$$

will be used. They follow from Lemma A.3 in Appendix A. The same appendix verifies

$$\frac{7}{25} < J_0(19/10) < \frac{283}{1000}, \qquad \frac{J_1(19/10)}{19/10} > \frac{3}{10}. \tag{13}$$

These use only the degree-six and degree-eight alternating Bessel bounds, recorded explicitly in the appendix.

Lemma 3.1. *One has $z > 0$ and $g > 0$ throughout $0 < r < k$.*

Proof. Set $W = p\phi' - \phi p'$. The two Bessel equations give

$$(rW)' = r(a_0 - r^{-2})p\phi.$$

Both endpoint limits of rW are zero. Its derivative is negative before $r_0 = a_0^{-1/2}$ and positive after r_0. Hence $W < 0$ in the open interval. Thus $f_r = W/p^2 < 0$ and $z = f_r/t' > 0$.

For the other sign put $\psi = -\phi' = \sqrt{\lambda} J_1(\sqrt{\lambda} r) > 0$. Now

$$[r(p\psi' - \psi p')]' = a_0 r p \psi > 0.$$

The expression on the left before differentiation has limit zero at the center. Consequently $h = \psi/p$ is strictly increasing. Using $\phi(r) = \int_r^k \psi(s)\,\mathrm{d}s$ gives

$$p\phi + u\phi' = p(r)\int_r^k p(s)[h(s) - h(r)]\,\mathrm{d}s > 0.$$

Since $g = (p\phi + u\phi')/S$, the claim follows. □

Put $m = a_0 - r^{-2}$ and $Q = a_0 u - c$.

Lemma 3.2. *For $0 < r < k$,*

$$g_r = \frac{u}{S}(pmg + Qz), \qquad z_r = -\frac{p}{u} g_r. \tag{14}$$

Proof. Use $u' = -p$, $p'' = -p'/r - (1 - r^{-2})p$ and $\phi'' = -\phi'/r - \lambda\phi$. First,

$$S' + S/r = p(u/r^2 - c).$$

Differentiating $z = (\phi p' - p\phi')/S$ in r and then dividing by $t' = -S/p^2$ gives

$$f_{tt} = \frac{p^3}{S^3}\left[\phi(p'^2 - pp'') + \phi'' S - \phi' S'\right].$$

After inserting $p' = u - c - p/r$, the bracket equals $S[a_0\phi - cz - (p/r^2)g]$. Hence

$$f_{tt} = \frac{p^3}{S^2}\left[a_0\phi - cz - \frac{p}{r^2}g\right].$$

Now use $g_r = -tz_r$, $z_r = f_{tt}t'$ and $\phi = pg + uz$ to obtain (14). □

Lemma 3.3. *There is a unique $r_Q \in (19/10, k)$ for which $Q(r_Q) = 0$. For*

$$F(r) = \frac{p(r)m(r)}{Q(r)}$$

one has $F'(r) > 0$ both on $(0, r_0)$ and on (r_Q, k).

Proof. Since $Q' = -a_0 p < 0$, $Q(k) = -c < 0$, and

$$Q(19/10) > \frac{3}{5}\frac{7}{25} - \frac{2}{5}\frac{41}{100} = \frac{1}{250} > 0,$$

the zero is unique and lies after 19/10. Also $r_0 \le \sqrt{5/3} < 19/10$. Write $Q_0 = Q(0) = a_0 - \lambda c > 2/5$.

For $0 < r < r_0$, put $x = a_0 r^2 \in (0, 1)$. The elementary Bessel bounds $p \le r/2$ and $p'/p \le 1/r$ hold there; the latter follows from $p' = p/r - J_2(r)$ and the direct alternating-series bound $J_2(r) \ge r^2/8 - r^4/96 > 0$ for $r < r_0 \le \sqrt{5/3}$. Furthermore

$$Q = Q_0 - a_0 \int_0^r p(s)\,\mathrm{d}s \ge Q_0 - x/4 > 0.$$

As $F < 0$, the condition $F' > 0$ is equivalent to

$$\frac{2}{r(1-x)} - \frac{p'}{p} > \frac{a_0 p}{Q}.$$

The left side is at least $(1 + x)/[r(1 - x)]$, while the right side is at most $x/[2r(Q_0 - x/4)]$. The required strict comparison follows from

$$2(Q_0 - x/4)(1 + x) - x(1 - x) = 2Q_0 + (2Q_0 - 3/2)x + x^2/2 > 4/5 - 7x/10 + x^2/2 \ge 1/10.$$

For $r > r_Q$ put $A = -p'/p$ and $\ell = r - r_Q$. The estimates (13) give $p'(19/10) < 283/1000 - 3/10 < 0$. Since $(\log p)'' < 0$ for $r > \sqrt{5}/2$, we have $A > 0$ on $[r_Q, k)$. The tangent-line bound for the concave function $\log p$ gives, for $r_Q \le s \le r$,

$$p(s) \le p(r)e^{A(r)(r-s)}, \qquad \frac{a_0 p(r)}{-Q(r)} = \frac{p(r)}{\int_{r_Q}^r p(s)\,\mathrm{d}s} \ge \frac{A(r)}{e^{A(r)\ell} - 1}.$$

It follows that

$$A + \frac{a_0 p}{-Q} \ge \frac{A}{1 - e^{-A\ell}} > \frac{1}{\ell}.$$

On the other hand,

$$\frac{m'}{m} = \frac{2}{r(a_0 r^2 - 1)} < \frac{1}{r - r_Q}.$$

Indeed, the last strict inequality is equivalent to $a_0 r^3 - 3r + 2r_Q > 0$; it holds at $r = r_Q > r_0$ and its derivative is positive thereafter. Therefore

$$\frac{\mathrm{d}}{\mathrm{d}r}\log(-F) = -A + \frac{m'}{m} - \frac{a_0 p}{-Q} < 0.$$

As $F < 0$ on this interval too, this proves $F' > 0$. □

Proposition 3.4. *For every $0 < r < k$,*

$$g_r > 0, \qquad f_{tt} > 0, \qquad 0 < g(r) < \sqrt{K}.$$

Proof. At any zero of g_r in an interval with $Q \neq 0$, the system (14) gives $z_r = 0$ and

$$g_{rr} = \frac{u}{S}\, gQ \left(\frac{pm}{Q}\right)'. \tag{15}$$

The endpoint expansions give

$$g_r \longrightarrow \frac{a_0 - \lambda c}{1 + c} > 0 \quad (r \downarrow 0),$$

and, with $\delta = k - r$,

$$g = \beta - \frac{f_2}{4}\delta^2 + O(\delta^3), \qquad f_2 = \frac{2\beta}{3}(a_0 - k^{-2}) > 0.$$

Thus $g_r > 0$ near both endpoints. On $(0, r_0)$, Lemma 3.3 and (15) make every hypothetical zero of g_r a strict crossing from negative to positive. Since g_r starts positive, no such zero is possible. On $[r_0, r_Q]$ both terms in $pmg + Qz$ are nonnegative and at least one is positive, so $g_r > 0$ directly. On (r_Q, k) every hypothetical zero is instead a strict crossing from positive to negative. Positivity near k excludes such a zero. This proves $g_r > 0$ globally.

Because $g_r = -tt' f_{tt}$ and $t > 0$, $t' < 0$, strict convexity follows. Finally $g(0+) = 0$ and $g(k-) = \beta$ give the claimed bound. □

4. A quadratic estimate

The elementary coarse bounds needed below are

$$\frac{1}{4} < \lambda < \frac{2}{5}, \qquad \frac{2}{5} < c < \frac{41}{100}, \qquad 2 < k < 4. \tag{16}$$

These are included in Lemma A.3; in particular, the lower bound on λ follows from $q > 2$ and $k < 4$.

4.1. **Elementary bounds for the phase.** Set

$$\rho = \sqrt{u^2 + p^2}, \qquad \theta = \arctan(p/u), \qquad \eta = \theta' = \frac{S}{\rho^2} > 0.$$

The phase increases from zero to $\pi/2$. Write $s_\theta = \sin\theta = p/\rho$ and $c_\theta = \cos\theta = u/\rho$. The system $u' = -p$, $p' = u - c - p/r$ gives

$$\eta = 1 - \frac{c}{\rho}c_\theta - \frac{s_\theta c_\theta}{r}, \qquad \frac{\rho'}{\rho} = -\frac{c}{\rho}s_\theta - \frac{s_\theta^2}{r}. \tag{17}$$

Lemma 4.1. *Throughout $0 < r < k$,*

$$\eta > \frac{s_\theta c_\theta}{r}, \qquad \eta < \frac{\tan\theta}{r}, \qquad \eta > \frac{1}{2} - \frac{s_\theta c_\theta}{r}. \tag{18}$$

Proof. First $J_2 > 0$ and $J_3 > 0$ on $(0, 4)$. For J_3, the first two alternating terms give $J_3(r) \geq r^3/48 - r^5/768 > 0$. For J_2, four terms give

$$J_2(r) \geq r^2\left(\frac{1}{8} - \frac{r^2}{96} + \frac{r^4}{3072} - \frac{r^6}{184320}\right) > 0.$$

Indeed the polynomial in parentheses, as a function of $x = r^2$, is decreasing on $[0, 16]$ and has value $7/360$ at 16.

Let $F = p^2 - uJ_2$. The recurrences $p' = p/r - J_2$ and $J_2' = p - 2J_2/r$ give the exact first-order equation

$$(r^2 F)' = rp(2p - cr).$$

Since $(p/r)' = -J_2/r < 0$, the forcing has exactly one change of sign, from positive to negative. Both endpoint limits of r^2F are zero. Consequently $F > 0$ in the open interval. It follows that $S = up/r + F > up/r$, proving the first bound in (18).

The recurrence $4J_2/r = p + J_3$ gives another exact identity:

$$p\rho^2 - ruS = p\left(p - \frac{ru}{2}\right)^2 + \frac{r^2u^2}{4}J_3 > 0.$$

Division by $ru\rho^2$ proves the second bound.

Finally $(J_0^2 + J_1^2)' = -2p^2/r < 0$ and $J_0(k)^2 + J_1(k)^2 = c^2$. Hence

$$\rho^2 - 2cu = J_0(r)^2 + p(r)^2 - c^2 > 0.$$

Thus $cu/\rho^2 < 1/2$, which combined with (17) gives the third bound. □

Lemma 4.2. *If $\theta \geq \pi/4$, then $r > 2$ and*

$$r > 2\sqrt{2}\sin\theta.$$

Proof. The elementary series bounds $J_0(2) > 11/50$, $J_1(2) < 3/5$, together with $c > 2/5$, give $u(2) > p(2)$. Thus $\theta(2) < \pi/4$ and $\theta \geq \pi/4$ implies $r > 2$.

By the second identity in the preceding proof,

$$\left(\log\frac{\sin\theta}{r}\right)' = \cot\theta\left(\eta - \frac{\tan\theta}{r}\right) \leq -\frac{ru^2J_3}{4p\rho^2} < 0. \tag{19}$$

On $0 < r \leq 2$ we have $u^2/\rho^2 > 1/2$, $p \leq r/2$ and

$$J_3(r) \geq \frac{r^3}{48} - \frac{r^5}{768} \geq \frac{r^3}{64}.$$

Therefore the right side of (19) is at most $-r^3/256$. Integrating from zero to 2, using $\lim_{r\downarrow 0}\sin\theta/r = [2(1+c)]^{-1} \leq 5/14$, gives

$$\frac{\sin\theta(2)}{2} < \frac{5}{14}e^{-1/64} < \frac{5}{14}\frac{64}{65} = \frac{32}{91} < \frac{1}{2\sqrt{2}}.$$

The last inequality is the integer comparison $8192 < 8281$. Monotonicity in (19) extends this bound to every $r \geq 2$. □

4.2. **Monotonicity of an auxiliary potential.**

Proposition 4.3. *The function*

$$\mathcal{V}(r) = -\frac{\rho'' + \rho'/r}{\rho}$$

is strictly decreasing on $(0, k)$.

Proof. Direct differentiation of (17) gives

$$\eta' = \frac{s_\theta c_\theta}{r^2} + (2\eta - 1)\frac{c}{\rho}s_\theta + \frac{\eta(2s_\theta^2 - 1)}{r},$$

$$\mathcal{V} = 1 - \frac{c}{\rho}c_\theta - \frac{s_\theta^2}{r^2} - \eta^2,$$

and consequently

$$\mathcal{V}' = -\frac{cp}{\rho^2}(1-2\eta)^2 + \frac{2}{r}\left[\left(\eta c_\theta - \frac{s_\theta}{r}\right)^2 - \eta^2 s_\theta^2\right]. \tag{20}$$

We show that the square bracket is strictly negative.

For $0 < \theta < \pi/4$ the bracket, as a quadratic in η, has positive roots

$$\eta_- = \frac{s_\theta}{r(s_\theta + c_\theta)}, \qquad \eta_+ = \frac{s_\theta}{r(c_\theta - s_\theta)}.$$

Lemma 4.1 gives

$$\eta > \frac{s_\theta c_\theta}{r} \geq \eta_-, \qquad \eta < \frac{s_\theta}{r c_\theta} < \eta_+.$$

Here $c_\theta(s_\theta + c_\theta) \geq 1$ is equivalent to $c_\theta \geq s_\theta$. At $\theta = \pi/4$ the bracket is linear in η and the same lower bound still makes it strictly negative.

For $\pi/4 < \theta < \pi/2$ the quadratic has negative leading coefficient and is negative whenever $\eta > \eta_-$. By Lemma 4.2, $r > 2\sqrt{2}s_\theta$. On this angular range

$$c_\theta + \frac{1}{s_\theta + c_\theta} \leq \sqrt{2}.$$

Indeed its value at $\pi/4$ is $\sqrt{2}$, and its derivative is $-s_\theta + (s_\theta - c_\theta)/(s_\theta + c_\theta)^2 < 0$. It follows that

$$r > 2s_\theta \left(c_\theta + \frac{1}{s_\theta + c_\theta} \right), \qquad \frac{1}{2} - \frac{s_\theta c_\theta}{r} > \frac{s_\theta}{r(s_\theta + c_\theta)} = \eta_-.$$

The third bound in (18) again gives $\eta > \eta_-$. Thus the bracket in (20) is negative everywhere, proving $\mathcal{V}' < 0$. □

4.3. **A Wronskian comparison.**

Proposition 4.4. *For every $t > 0$,*

$$g^2 + z^2 < K.$$

Proof. The endpoint expansions give

$$\mathcal{V}(0+) = \frac{c + 1/2}{(1+c)^2} > \frac{2}{5} > \lambda, \qquad \mathcal{V}(k-) = \frac{1}{4} - \frac{1}{k^2} < \frac{1}{4} < \lambda.$$

For the first strict comparison, a sufficient rational bound is $\mathcal{V}(0+) > (9/10)/(141/100)^2 > 2/5$. By Proposition 4.3, $\mathcal{V} - \lambda$ has exactly one sign change, from positive to negative.

Let $W = \rho\phi' - \phi\rho'$. Since $\phi'' + \phi'/r + \lambda\phi = 0$,

$$(rW)' = r\rho\phi(\mathcal{V} - \lambda).$$

Both endpoint limits of rW are zero. The forcing is first positive and then negative, so $W > 0$ throughout $(0, k)$. Using $\rho' = -p(c + p/r)/\rho$ and the formulas for z, g gives the exact relation

$$z - tg = -\frac{\rho W}{pS} < 0.$$

Finally $g_t = -tf_{tt}$ and $z_t = f_{tt}$ imply

$$\frac{\mathrm{d}}{\mathrm{d}t}(g^2 + z^2) = 2f_{tt}(z - tg) < 0.$$

As $t \downarrow 0$, $(g, z) \to (\beta, 0)$, giving the energy bound. □

5. A lower bound for the first derivative

In this section put $v = 1 - J_0(r)$ and $\sigma = 1 + c = u + v$. We first need an estimate for the disk profile.

Lemma 5.1. *On $0 < r < k$,*

$$\frac{2vS}{p^2} < \sigma. \tag{21}$$

Proof. The standard Bessel product and derivative recurrence give

$$J_1(r) = \frac{r}{2}\prod_{n\geq 1}\left(1 - \frac{r^2}{j_{1,n}^2}\right), \qquad \frac{J_2(r)}{rJ_1(r)} = \sum_{n\geq 1}\frac{2}{j_{1,n}^2 - r^2}.$$

The series and its derivative converge uniformly on each compact subinterval of $(0, k)$. In particular $J_2/(rJ_1)$ is strictly increasing there. See [14, Eqs. 10.21.15 and 10.6.2].

Since $X = J_0(r)$ is strictly decreasing from 1 to $-c$, regard $E(X) = p(r)^2$ as a function of $X \in [-c, 1]$. The function E is continuous on the closed interval, vanishes at its endpoints and is smooth in its interior. Using $X' = -p$ and $p' = X - p/r$ gives

$$E_X = -2X + \frac{2p}{r}, \qquad E_{XX} = -2 + \frac{2J_2(r)}{rp}, \qquad E_{XXX} < 0.$$

The last sign follows because $J_2/(rp)$ increases with r while X decreases with r.

For fixed endpoints $x_- = -c$, $x_+ = 1$, the second divided difference is

$$E[x_-, X, x_+] = -\frac{E(X)}{(x_+ - X)(X - x_-)}.$$

Its derivative with respect to the middle node is the third divided difference $E[x_-, X, X, x_+]$. The Hermite divided-difference mean-value theorem yields, for some $\xi \in (x_-, x_+)$,

$$\frac{d}{dX}E[x_-, X, x_+] = E[x_-, X, X, x_+] = \frac{E_{XXX}(\xi)}{3!} < 0.$$

Only continuity of E at x_-, x_+ is needed here: the usual proof subtracts its cubic Hermite interpolant and applies Rolle's theorem three times to interior points. No endpoint second derivative is being assumed.

It follows that $E(X)/[(1 - X)(X + c)]$ is strictly increasing with X, and hence strictly decreasing with r. Differentiating that ratio gives precisely

$$\left(\frac{p^2}{uv}\right)' = -\frac{p}{u^2v^2}\{(u - v)p^2 - 2uvp'\} < 0.$$

Finally $\sigma p^2 - 2vS = (u - v)p^2 - 2uvp'$, proving (21). □

Corollary 5.2. *Put $h_0 = (p + vA)/\sigma$. The function*

$$\mathcal{R} = \lambda - \frac{h_0}{r} + h_0^2 - h_0'$$

is strictly decreasing on $(0, k)$.

Proof. Logarithmic differentiation of the same Bessel product gives $A' > 0$. Also p/r is strictly decreasing, so $rp < 2\int_0^r p(s)\,ds = 2v$. Direct differentiation, using $h_0' = vA'/\sigma$, gives

$$\mathcal{R}' = \frac{pA'}{\sigma}\left(\frac{2vS/p^2}{\sigma} - 1\right) + \frac{rp - 2v}{\sigma r^3} < 0$$

by Lemma 5.1. □

We will also use $J_0(qs) > 1 - s$ for $0 < s < 1$. Here is an elementary proof. J_1 is strictly unimodal on $(0, q)$: its derivative is positive for $r \le \sqrt{5}/2$ by the change-of-variable lemma, and at any later zero its second derivative is $-(1 - r^{-2})J_1$, so a downward crossing can occur only after 1, and all such later crossings are downward. The elementary series gives $J_1'(1) > 5/16 > 0$, and $J_1'(q) = -J_1(q)/q < 0$, so there is exactly one such crossing. Moreover $qJ_1(q) > 1$. Indeed $2 < q < 5/2$ and the four-term alternating lower bound gives, with $s = q^2 \in [4, 25/4]$,

$$qJ_1(q) \ge Q(s), \quad Q(s) = \frac{s}{2} - \frac{s^2}{16} + \frac{s^3}{384} - \frac{s^4}{18432}.$$

Here $Q''(s) = -[(s-12)^2+48]/1536 < 0$ and $Q(4) = 83/72 > 1$, $Q(25/4) = 5834975/4718592 > 1$. Therefore $J_1(r) - 1/q$ has exactly one sign change on $(0, q)$, from negative to positive. The function $J_0(r) - 1 + r/q$ first increases and then decreases, with zero values at both endpoints, which proves the assertion.

Proposition 5.3. *For $0 < r < k$,*

$$z(r) > \frac{k - r}{k(1 + c)}.$$

Proof. Use h_0 from the preceding corollary. Let ψ be the positive solution of

$$\psi' = -h_0\psi, \qquad \psi(0) = 1.$$

At the center $h_0 = r/(4\sigma) + O(r^3)$, so ψ extends as a smooth even positive function. At the other endpoint, with $\delta = k - r$, the elementary Bessel expansions give

$$p = c\delta + \frac{c}{2k}\delta^2 + O(\delta^3), \quad v = \sigma - \frac{c}{2}\delta^2 + O(\delta^3), \quad h_0 = \frac{1}{\delta} + \frac{1}{2k} + O(\delta).$$

Consequently $\psi = c_\psi\delta(1 + \delta/(2k) + O(\delta^2))$ for some $c_\psi > 0$. In particular ψ and ϕ have simple zeros at k.

Define

$$V_\psi = -\frac{\psi'' + \psi'/r}{\psi} = h_0' + h_0/r - h_0^2 = \lambda - \mathcal{R}.$$

By the preceding corollary V_ψ is strictly increasing. For $W = \psi\phi' - \phi\psi'$ one has the exact identity

$$(rW)' = r\psi\phi(V_\psi - \lambda).$$

The central regularity and the simple boundary zeros give $rW \to 0$ at both endpoints. The strictly increasing function $V_\psi - \lambda$ must therefore change sign exactly once, from negative to positive: otherwise its integral against the positive weight $r\psi\phi$ could not be zero. Thus rW first decreases from zero and then increases to zero, and $W < 0$ throughout the open interval. It follows that

$$-\frac{\phi'}{\phi} - h_0 = -\frac{W}{\psi\phi} > 0.$$

Finally the exact transformed derivative is $z = (\phi p' - p\phi')/S$. Direct subtraction, using $\sigma - u = v$, yields

$$z - \frac{\phi}{\sigma} = \frac{\phi p}{S}\left(-\frac{\phi'}{\phi} - \frac{p - vp'/p}{\sigma}\right) = \frac{\phi p}{S}\left(-\frac{\phi'}{\phi} - h_0\right) > 0.$$

The inequality for J_0 proved above, rescaled to $(0, k)$, gives $\phi(r) > (k - r)/k$ and completes the proof. □

6. Concavity of the auxiliary function

Write $q_n = j_{0,n}$ and $\kappa_n = j_{1,n}$.

Lemma 6.1. *For every $n \geq 1$ and every $x \geq q_n$,*

$$B(\kappa_n) \leq B(x) \leq B(q_n).$$

Equality on the left and right holds only at $x = \kappa_n$ and $x = q_n$, respectively.

Proof. One has $B'(x) = J_0(x)J_1(x)/x$. By interlacing, the positive zeros q_n of J_0 give local maxima of B and the positive zeros κ_n of J_1 give local minima. The identities

$$2B(x) - \frac{J_1(x)^2}{x} = 3\int_0^x \frac{J_1(s)^2}{s^2} ds$$

and

$$B(q_{n+1}) - B(q_n) = -\frac{1}{2}\int_{q_n}^{q_{n+1}} \frac{J_0(s)^2}{s^2} ds < 0$$

show, respectively, that the successive minima increase strictly and the successive maxima decrease strictly. Between successive critical points B is strictly monotone. The asserted bounds follow for each n. Also $B(q) > B(k) > 0$ and $K = \alpha B(q)/B(k)$. □

Put $L(r) = KB(r) - \alpha B(\alpha r)$, so that $G(r) = K - rL(r)/p(r)^2$. The endpoint identities give $G(k) = 0$ and

$$G_r(k) = -\frac{2K}{3k}. \tag{22}$$

The singularities at $0, k, -k$ are removable. The function G is an even meromorphic function whose only other possible poles are double poles at $\pm\kappa_n$, $n \geq 2$. There are no unlisted complex zeros: if $J_1(\zeta) = 0$ with $\zeta \neq 0$, multiply the angular-one radial Dirichlet equation for $v(s) = J_1(\zeta s)$ on $(0, 1)$ by $\bar{v}$ to get

$$\int_0^1 \left(s|v'|^2 + \frac{|v|^2}{s} \right) ds = \zeta^2 \int_0^1 s|v|^2 \, ds.$$

The two integrals are strictly positive, so $\zeta^2 > 0$ and ζ is real. The nonzero zeros are simple by uniqueness for the Bessel ODE.

The fixed bounds needed in the following lemma, including $K > .654$, $B(\kappa_3) > .634$, $B(q_2) < .640$ and $J_1(q_2)^2 < .116$, are proved in Lemma A.3.

Lemma 6.2. *For $n \geq 2$, $L(\kappa_n) > 0$. Moreover*

$$3L(\kappa_n) - \phi(\kappa_n)\phi_r(\kappa_n) > 0.$$

Proof. Lemma 6.1 gives $B(x) \leq B(q)$ for $x \geq q$ and $B(\kappa_n) > B(k)$ for $n \geq 2$. Since $KB(k) = \alpha B(q)$, this proves $L(\kappa_n) > 0$.

For $n = 2$, the inequalities $k < \alpha\kappa_2 < q_2$ imply $\phi(\kappa_2) < 0$ and $\phi_r(\kappa_2) > 0$, so the second claim is immediate. For $n \geq 3$ we have $B(\kappa_n) \geq B(\kappa_3) > .634$ and $B(\alpha\kappa_n) \leq B(q_2) < .640$. Also $E(x) = J_0(x)^2 + J_1(x)^2$ satisfies $E' = -2J_1(x)^2/x \leq 0$. Thus for $x = \alpha\kappa_n \geq q_2$,

$$|\phi(\kappa_n)\phi_r(\kappa_n)| = \alpha|J_0(x)J_1(x)| \leq \frac{\alpha}{2}E(x) < \frac{.628 \cdot .116}{2}.$$

The coarse bounds now give the strictly positive rational estimate

$$3L(\kappa_n) - \phi(\kappa_n)\phi_r(\kappa_n) > 3(.654 \cdot .634 - .628 \cdot .640) - \frac{.628 \cdot .116}{2} = \frac{431}{250000} > 0.$$

□

6.1. **Uniform complex asymptotics and coefficient residues.** For the coefficient calculation we use the classical uniform asymptotic formula, for $\nu = 0, 1$,

$$J_\nu(\zeta) = \left(\frac{2}{\pi\zeta}\right)^{1/2} \left[\cos\left(\zeta - \frac{\nu\pi}{2} - \frac{\pi}{4}\right) + O\left(\frac{e^{|\operatorname{Im}\zeta|}}{|\zeta|}\right)\right], \qquad |\arg\zeta| \le \frac{\pi}{2}. \tag{23}$$

The error is uniform on the closed sector, with the principal square root. This follows from the Hankel-function remainder estimates in [14, §10.17(iv), Eqs. 10.17.13–10.17.15] and $2J_\nu = H_\nu^{(1)} + H_\nu^{(2)}$; the variation bound is at most $\pi/(2|\zeta|)$ on the relevant half-plane, and the conjugate bound handles the other Hankel function. Integer-order parity extends the absolute-value estimates to the left half-plane.

Lemma 6.3. *For $R_N = (N + 3/4)\pi$ and all sufficiently large integers N, G has no pole on $|\zeta| = R_N$ and*

$$\sup_{|\zeta|=R_N} |G(\zeta)| \le CR_N^2,$$

where C is independent of N.

Proof. Write $R = R_N$ and $\zeta = x + iy$. By parity assume $x \ge 0$. If $|y| \ge 1$, then

$$|\cos(\zeta - 3\pi/4)| \ge \sinh|y| \ge c_0 e^{|y|}$$

for a fixed $c_0 > 0$. If $|y| < 1$, then $0 \le R - x = y^2/(R + x) \le 1/R$ and $R - 3\pi/4 = N\pi$, so $|\cos(\zeta - 3\pi/4)| \ge \cos(1/R)$. Thus (23), with its error absorbed for large R, gives

$$|J_1(\zeta)| \ge c_1 R^{-1/2} e^{|y|}.$$

The same asymptotic formula gives

$$|J_0(\zeta)| + |J_1(\zeta)| \le C_1 R^{-1/2} e^{|y|}, \qquad |J_0(\alpha\zeta)| + |J_1(\alpha\zeta)| \le C_2 R^{-1/2} e^{\alpha|y|}.$$

Here α is fixed and positive. Consequently $|B(\zeta)| + |B(\alpha\zeta)| \le C_3 e^{2|y|}$. The representation $G = K - \zeta L/J_1^2$ now proves the claim. □

For a function analytic at zero, $[r^j]G$ denotes the coefficient of r^j in its Taylor series.

Lemma 6.4. *For every integer $m \ge 2$,*

$$-[r^{2m}]G(r) = \sum_{n=2}^{\infty} \frac{2[(2m-1)L(\kappa_n) - \phi(\kappa_n)\phi_r(\kappa_n)]}{\kappa_n^{2m+1} J_0(\kappa_n)^2}. \tag{24}$$

The series is absolutely convergent.

Proof. At $\kappa = \kappa_n$ let $x_\kappa = J_0(\kappa) \ne 0$. The Bessel equation gives

$$p(r) = x_\kappa(r - \kappa) - \frac{x_\kappa}{2\kappa}(r - \kappa)^2 + O((r - \kappa)^3).$$

Therefore the principal part of $K - G = rL/p^2$ at κ is

$$\frac{A_\kappa}{(r - \kappa)^2} + \frac{D_\kappa}{r - \kappa}, \qquad A_\kappa = \frac{\kappa L(\kappa)}{x_\kappa^2}, \quad D_\kappa = \frac{2L(\kappa) + \kappa L'(\kappa)}{x_\kappa^2}.$$

Since $B'(\kappa) = 0$, $\kappa L'(\kappa) = \phi(\kappa)\phi_r(\kappa)$. Evenness determines the principal part at $-\kappa$.

To justify the coefficient identity, apply the residue theorem to $G(\zeta)/\zeta^{2m+1}$ on $|\zeta| = R_N$, using Lemma 6.3. For each fixed integer $m \ge 2$,

$$\left|\frac{1}{2\pi i}\int_{|\zeta|=R_N} \frac{G(\zeta)}{\zeta^{2m+1}}\,\mathrm{d}\zeta\right| \le C_{\alpha,K} R_N^{2-2m} \longrightarrow 0. \tag{25}$$

The residue at zero is $[\zeta^{2m}]G(\zeta)$. With the principal parts already computed, the sum of the residues at κ and $-\kappa$ is

$$\frac{2(2m+1)A_\kappa}{\kappa^{2m+2}} - \frac{2D_\kappa}{\kappa^{2m+1}} = \frac{2[(2m-1)L(\kappa) - \phi(\kappa)\phi_r(\kappa)]}{\kappa^{2m+1}J_0(\kappa)^2}.$$

It remains to verify absolute convergence before taking the limit of the finite residue sums. The zero asymptotic [14, Eq. 10.21.19] gives

$$\kappa_n = (n + 1/4)\pi + O(n^{-1}).$$

Apply (23) on the positive axis at κ_n to obtain

$$J_0(\kappa_n) = (-1)^n \left(\frac{2}{\pi\kappa_n}\right)^{1/2} (1 + O(\kappa_n^{-1})).$$

The same upper bounds give $B(x) = O(1)$ as $x \to +\infty$, so $L(\kappa_n) = O(1)$ and

$$\phi(\kappa_n)\phi_r(\kappa_n) = -\alpha J_0(\alpha\kappa_n)J_1(\alpha\kappa_n) = O(\kappa_n^{-1}).$$

Consequently, for each fixed $m \geq 2$, the absolute value of the displayed paired residue is $O_m(\kappa_n^{-2m})$, and the resulting series converges absolutely. Letting $N \to \infty$ in the residue theorem and using (25) gives

$$-[\zeta^{2m}]G(\zeta) = \sum_{n=2}^{\infty} \frac{2[(2m-1)L(\kappa_n) - \phi(\kappa_n)\phi_r(\kappa_n)]}{\kappa_n^{2m+1}J_0(\kappa_n)^2}.$$

□

6.2. **Concavity and the final estimates.**

Proposition 6.5. *There are numbers $a_m > 0$ such that*

$$G(r) = 2\lambda - K - \sum_{m=1}^{\infty} a_m r^{2m}, \qquad |r| < \kappa_2.$$

For $0 < r < k$,

$$G(r) > 0, \qquad G_r(r) < 0, \qquad G_{rr}(r) < 0, \qquad G_t(t) > 0.$$

Proof. Lemma 6.2 makes every summand of (24) positive for $m \geq 2$. For $m = 1$, direct expansion at zero gives

$$[r^2]G = \frac{2\lambda - \lambda^2 - K}{4} < 0,$$

because $0 < \lambda < .4$ and $K > .654 > .64$. There are no singularities in $|r| < \kappa_2$, so its Taylor series converges there. Termwise differentiation proves both strict derivative signs. Since $G(k) = 0$, positivity follows. Finally $t_r < 0$ implies $G_t = G_r/t_r > 0$. □

Corollary 6.6. *For $\delta = k - r$, $0 \leq r \leq k$,*

$$0 \leq G(r) \leq \frac{2K}{3k}\delta, \qquad 0 \leq -G_r(r) \leq \frac{2K}{3k}.$$

Proof. Strict concavity and $G_r(0) = 0$ show that G_r decreases to the value in (22). A concave function lies below its tangent line at k, where $G(k) = 0$. □

Lemma 6.7. *For $0 < r < k$,*

$$0 < \frac{up}{S} < \frac{k^2 - r^2}{r}.$$

Proof. Write $b = k^2$, $x = r^2$, $p = rh(x)$ and use the canonical product

$$h(x) = \frac{1}{2}(1 - x/b)P(x), \qquad P(x) = \prod_{n\geq 2}(1 - x/\kappa_n^2).$$

Its logarithmic derivative is

$$\frac{h'}{h} = -\frac{1}{b-x} - \Sigma(x), \qquad \Sigma(x) = \sum_{n\geq 2}\frac{1}{\kappa_n^2 - x} \leq \Sigma(b) = \frac{1}{b}.$$

The final equality follows without a sum evaluation: the endpoint ODE $p''(k)/p'(k) = -1/k$, applied to the displayed factored product, gives $P'(b)/P(b) = -1/b$. Since P decreases, $h(y) \leq h(x)(b-y)/(b-x)$ for $x \leq y \leq b$. Consequently

$$u = \frac{1}{2}\int_x^b h(y)dy \leq \frac{b-x}{4}h(x), \qquad S = xh^2 + u(h + 2xh').$$

The claimed upper bound is equivalent to positivity of

$$J = (b-x)xh^2 + uD, \qquad D = (b-2x)h + 2x(b-x)h'.$$

If $D \geq 0$ this is immediate. If $D < 0$, the upper bound on u gives

$$J \geq \frac{(b-x)h^2}{4}[b - 2x(b-x)\Sigma(x)] \geq \frac{(b-x)h^2}{4}\left[b - \frac{2x(b-x)}{b}\right] > 0,$$

since $x(b-x) \leq b^2/4$. □

Corollary 6.8. *On $0 < r < k$,*

$$0 < d(r) < 2G(r) \leq \frac{4K}{3k}(k-r).$$

Proof. Since $t_r = -S/p^2$, $d = 2G + C_{\mathrm{rad}}G_r$ with $C_{\mathrm{rad}} = up/S$. The positive-coefficient expansion and $G(k) = 0$ give

$$d = 2\sum_{m\geq 1} a_m[(k^{2m} - r^{2m}) - mC_{\mathrm{rad}}r^{2m-1}].$$

Lemma 6.7 shows that each bracket is strictly positive, because

$$k^{2m} - r^{2m} = (k^2 - r^2)\sum_{j=0}^{m-1} k^{2(m-1-j)}r^{2j} \geq m(k^2 - r^2)r^{2m-2} > mC_{\mathrm{rad}}r^{2m-1}.$$

This proves $d > 0$. The strict upper bound $d < 2G$ follows from $tG_t > 0$, and the final linear estimate was proved above. □

6.3. **Completion of the matrix estimate.** The preceding sections give, on $0 < r < k$,

$$g^2 + z^2 < K, \qquad 0 < d < 2G \leq \frac{4K}{3k}(k-r), \qquad z > \frac{k-r}{k(1+c)}.$$

Since $G|_{r=0} = 2\lambda - K > 0$, one has $K < 2\lambda < 4/5 < 1$, and $c < 1/2$. It follows that $2\sqrt{K}(1+c) < 3$. Therefore

$$0 < d < \frac{4K}{3k}(k-r) < 2\sqrt{K}\frac{k-r}{k(1+c)} < 2\sqrt{K}\, z.$$

Together with the quadratic bound for g and z this proves

$$\boxed{d^2 < 4Kz^2 < 4K(K - g^2), \qquad 0 < r < k.} \tag{26}$$

By (26), the matrix M and the separate mixed-Hessian coefficient are positive semidefinite by Proposition 2.3 and Proposition 2.4. In particular, (11) holds throughout the parameter interval.

7. The functional inequality

We first record the endpoint control needed to apply the profiles to arbitrary smooth functions, including those that change sign. We write $\|\cdot\|_2$ for the L^2 norm on the domain under consideration and use the Frobenius norm for matrices.

Lemma 7.1. *With $\ell = (1+c)^{-1}$ and $\beta = \sqrt{K}$, their endpoint behavior is*

$$\begin{aligned} &f(t) = \beta + O(t^2), && b(t) = O(t), && G(t) = \frac{4K}{3k}t + O(t^2) && (t \downarrow 0),\\ &f(t) = \ell t + O(t^{-1}), && b(t) = 2\lambda\ell t + O(t^{-1}), && G(t) = 2\lambda - K + O(t^{-2}) && (t \to \infty). \end{aligned}$$

The expansions are convergent analytic or Laurent expansions at the respective endpoints. The functions

$$f_t, \quad f - tf_t, \quad b_t, \quad b - tb_t, \quad G, \quad G_t, \quad tG_t$$

are bounded on $(0,\infty)$.

Proof. At $r = k$, the expansions in Section 2.5 show that $t = \delta/2 + O(\delta^2)$ has an analytic inverse and $f = \beta + O(\delta^2)$. The formula for G gives the stated expansion. Moreover

$$A = \delta^{-1} + \frac{1}{2k} + O(\delta), \qquad fw = -K\delta^{-1} - \frac{K}{2k} + O(\delta).$$

In $b = -KA + G/r - fw$, both the pole and constant terms cancel. Hence $b = O(\delta)$, and all three functions are analytic in t at zero.

At $r = 0$, the function $y = 1/t = p/u$ is odd and analytic, with $y_r(0) = 1/[2(1+c)]$. Its inverse $r = r(y)$ is odd and analytic. The functions $f = \phi/p$ and b are odd meromorphic functions of r with at most a simple pole, while G is even and analytic there. The leading terms are

$$t = \frac{2(1+c)}{r} + O(r), \qquad f = \frac{2}{r} + O(r), \qquad b = \frac{4\lambda}{r} + O(r), \qquad G = 2\lambda - K + O(r^2).$$

Substitution of $r = r(1/t)$ proves the expansions at infinity and the claim about differentiated remainders. The boundedness assertions follow at the endpoints from these expansions and on compact subintervals from smoothness. □

Extend f and G evenly and b oddly to the real line. The even extension of G is Lipschitz, not differentiable at zero; only its first weak derivative will be used. Put

$$\mathcal{F}(a,P) = \begin{cases} Pf(a/P), & P > 0,\\ \ell|a|, & P = 0, \end{cases} \qquad \ell = (1+c)^{-1}.$$

The following is the functional form of the comparison.

Proposition 7.2. *For every real $U \in C_c^\infty(\mathbb{R}^2)$, the function $F = \mathcal{F}(U, |\nabla U|)$ is compactly supported and Lipschitz, and*

$$\int_{\mathbb{R}^2} |\nabla F|^2 - \lambda \int_{\mathbb{R}^2} F^2 \le K\left(\int_{\mathbb{R}^2} |\Delta U|^2 - \int_{\mathbb{R}^2} |\nabla U|^2\right). \tag{27}$$

Proof. On $\{P := |\nabla U| > 0\}$ choose the orthonormal frame $e_1 = \nabla U/P$, $e_2 = (-\partial_2 U, \partial_1 U)/P$, and write

$$X = e_1^T D^2 U e_1, \qquad Y = e_2^T D^2 U e_2, \qquad Z = e_1^T D^2 U e_2, \qquad s = U/P.$$

In the following identities all profiles are evaluated at s. The chain rule gives

$$|\nabla F|^2 = (zP + gX)^2 + g^2 Z^2.$$

For a 2×2 matrix, cof denotes its cofactor matrix. Direct calculation, or the identity $\operatorname{div}(\operatorname{cof} D^2 U) = 0$, gives

$$\begin{aligned} \operatorname{div}\big(Pb(s)\nabla U\big) &= P\big((2b - sb_s)X + bY\big) + b_s P^2, \\ \operatorname{div}\big(G(s)\operatorname{cof}(D^2 U)\nabla U\big) &= (2G - sG_s)(XY - Z^2) + PG_s Y. \end{aligned}$$

Adding these two expressions to

$$K(|D^2 U|^2 - P^2) - (|\nabla F|^2 - \lambda F^2)$$

produces exactly

$$(X, Y, P)M(s)(X, Y, P)^T + (2K - g^2 - d)Z^2.$$

For $s > 0$ this is nonnegative by the matrix estimate proved in Section 6. For $s < 0$, the parity extensions give

$$M(-s) = DM(s)D, \qquad D = \operatorname{diag}(-1, -1, 1),$$

and the same conclusion follows.

It remains to justify the calculation across the nodal and critical sets. If a scalar profile a has bounded a_s and $a - sa_s$, then $Pa(A/P)$ has bounded derivatives in (A, P) for $P > 0$. Integrating these bounds separately in A and P gives a Lipschitz extension to $\mathbb{R} \times [0, \infty)$. Lemma 7.1 applies this observation to f and b. Their extensions at $P = 0$ are $\ell|A|$ and $2\lambda\ell A$, respectively. Therefore F and

$$V_1 = Pb(U/P)\nabla U$$

with this extension are compactly supported Lipschitz functions.

For the second vector field, define

$$\Psi(A, \xi) = G(A/|\xi|)\xi \quad (\xi \neq 0), \qquad \Psi(A, 0) = 0.$$

Its derivative in A has norm at most $\|G_s\|_\infty$ and its matrix derivative in ξ is

$$GI - sG_s \frac{\xi \otimes \xi}{|\xi|^2}, \qquad s = A/|\xi|.$$

The bounds on G, G_s, sG_s prove that Ψ is globally Lipschitz; continuity at $\xi = 0$ follows from boundedness of G. Thus

$$V_2 = \operatorname{cof}(D^2 U)\Psi(U, \nabla U)$$

is also compactly supported and Lipschitz. Both divergences are integrable weak derivatives and have integral zero. In particular, the cusp in G introduces no measure supported on $\{U = 0\}$.

The portion of $\{U = 0\}$ where $P > 0$ is locally a smooth curve and has area zero. On $\{P = 0\}$ one has $D^2 U = 0$ almost everywhere: apply the implicit function theorem to each first derivative of U to discard the zero sets on which its gradient is nonzero. At each remaining critical point, the derivative of $x \mapsto (U(x), \nabla U(x))$ is zero. A Lipschitz composition then has derivative zero. In V_2, the term differentiating the cofactor also vanishes because $\Psi(U, 0) = 0$. Hence, almost everywhere on $\{P = 0\}$,

$$\nabla F = 0, \qquad \operatorname{div} V_1 = \operatorname{div} V_2 = 0.$$

These assertions agree with the weak derivatives by Rademacher's theorem; see [7, §§3.1, 4.2]. The remaining density at such points is $\lambda\ell^2U^2 \geq 0$, including when the critical set has positive area. Integration, together with $\int |D^2U|^2 = \int |\Delta U|^2$, proves (27). □

We now prove Theorem 1.2.

Proof. Let $0 \neq v \in C_c^\infty(\Omega)$ and set

$$Q = \frac{\|\Delta v\|_2^2}{\|\nabla v\|_2^2}, \qquad \widetilde{\Omega} = \sqrt{Q}\,\Omega, \qquad U(x) = v(x/\sqrt{Q}).$$

In dimension two, $\|\Delta U\|_2^2 = \|\nabla U\|_2^2$. The function F in Proposition 7.2 is Lipschitz and has compact support in $\widetilde{\Omega}$, so it belongs to $H_0^1(\widetilde{\Omega})$ by mollification. At a nonzero maximum point of $|U|$, one has $\nabla U = 0$ and $F = \ell|U| > 0$; therefore F is not identically zero. Applying (27) gives

$$Q^{-1}\lambda_1^{\mathrm{D}}(\Omega) = \lambda_1^{\mathrm{D}}(\widetilde{\Omega}) \leq \lambda = \frac{j_{0,1}^2}{j_{1,1}^2}.$$

Taking the infimum over v and then using density in $H_0^2(\Omega)$ proves (1). The disk eigenvalues show that the constant is optimal. The proof only uses compactly supported tests, so boundedness of Ω is not required. □

8. Equality

We first isolate the distinction between equality almost everywhere and equality of open sets.

Lemma 8.1. *For every nonempty proper open subset W of a planar disk B,*

$$\Lambda_1(W) > \Lambda_1(B).$$

Proof. Choose $x_0 \in B \setminus W$. By Lemma B.1, the buckling minimum on W is attained, and every element of $H_0^2(W)$, extended by zero, has a continuous representative vanishing at x_0. If the two eigenvalues were equal, a minimizer on W would also minimize on B. Lemma B.2 identifies the first eigenspace on B as one-dimensional, spanned by a function strictly positive inside B. No nonzero element of this eigenspace vanishes at x_0. □

We now prove Corollary 1.3.

Proof. The Faber–Krahn inequality [8, 11] gives

$$|\Omega|\lambda_1^{\mathrm{D}}(\Omega) \geq \pi q^2$$

for every open set of finite positive area. Multiplication by k^2/q^2 and Theorem 1.2 give (2).

If equality holds in (2), then equality holds in the Faber–Krahn inequality. Its equality characterization gives a disk B of area $|\Omega|$ such that Ω and B agree almost everywhere. Openness now implies $\Omega \subset B$, since an open neighborhood of any point outside B, including a point on its boundary, has positive area outside B. The equality of areas gives $\Lambda_1(\Omega) = \Lambda_1(B)$. Lemma 8.1 excludes a proper inclusion. Conversely, a disk gives equality by its explicit first eigenfunction. □

Remark 8.2. For a smooth bounded simply connected planar domain, the first Dirichlet Stokes eigenvalue equals $\Lambda_1(\Omega)$ under the stream-function representation; see [10, Remark 1]. Here the Stokes eigenvalue is the infimum of $\int_\Omega |\nabla v|^2 / \int_\Omega |v|^2$ over nonzero $v \in H_0^1(\Omega;\mathbb{R}^2)$ with $\operatorname{div} v = 0$. Thus the area inequality applies to that problem in the simply connected class. On a multiply connected domain, a stream function can have different constants on different

boundary components and need not belong to $H_0^2(\Omega)$. The present argument does not remove that distinction, nor does it prove the buckling conjecture in dimensions $N \geq 3$.

Appendix A. Bessel constants and root identification

The estimates in this appendix are finite comparisons of explicitly stated rational polynomials and their alternating remainders.

Lemma A.1. *If $0 < a < b$ are distinct positive zeros of J_1, then $b - a > \pi$.*

Proof. The function $y(r) = \sqrt{r}\, J_1(r)$ satisfies

$$y'' + \left(1 - \frac{3}{4r^2}\right) y = 0 \qquad (r > 0).$$

It is a nonzero member of $H_0^1(a, b)$. Multiplication by y and integration by parts, followed by the one-dimensional Dirichlet Wirtinger inequality, gives

$$\frac{\pi^2}{(b-a)^2} \int_a^b y^2\, dr \leq \int_a^b (y')^2\, dr = \int_a^b \left(1 - \frac{3}{4r^2}\right) y^2\, dr < \int_a^b y^2\, dr.$$

Since the last integral is positive, $b - a > \pi$. □

Lemma A.2. *Writing $q_n = j_{0,n}$ and $\kappa_n = j_{1,n}$, one has*

zero	*rational interval*	*endpoint signs*
$q = q_1$	$(2.40482, 2.40483)$	$(+, -)$
$k = \kappa_1$	$(3.83170, 3.83171)$	$(+, -)$
q_2	$(5.5200, 5.5202)$	$(-, +)$
κ_2	$(7.015, 7.016)$	$(-, +)$
κ_3	$(10.173, 10.174)$	$(+, -)$.

(28)

Each displayed interval contains exactly its indicated zero.

Proof. Here and below, the signs for q_n concern J_0 and those for κ_n concern J_1. For $\nu = 0, 1$ put

$$S_{\nu,N}(x) = \sum_{m=0}^{N} \frac{(-1)^m (x/2)^{2m+\nu}}{m!(m+\nu)!}.$$

For $0 < x < 11$, the absolute terms in the tail starting at $m = 31$ strictly decrease: their successive ratio is

$$\frac{x^2}{4(m+1)(m+\nu+1)} < \frac{121}{4 \cdot 32^2} < 1 \qquad (m \geq 31).$$

Consequently

$$S_{\nu,31}(x) < J_\nu(x) < S_{\nu,30}(x), \qquad 0 < S_{\nu,30}(x) - J_\nu(x) < \frac{(x/2)^{62+\nu}}{31!(31+\nu)!}. \tag{29}$$

The following table gives integers A, B for which

$$\frac{A}{10^8} < S_{\nu,31}(x) < S_{\nu,30}(x) < \frac{B}{10^8}.$$

Every entry is a rational comparison obtained by substitution in the two displayed finite sums and multiplication by positive denominators.

ν	x	A	B
0	2.40482	288	289
0	2.40483	−231	−230
1	3.83170	240	241
1	3.83171	−163	−162
0	5.5200	−2658	−2657
0	5.5202	4147	4148
1	7.015	−17608	−17607
1	7.016	12404	12405
1	10.173	11689	11690
1	10.174	−13281	−13280
1	2.40482	51914869	51914870
0	3.83170	−40275940	−40275939
1	5.5201	−34026346	−34026345
0	10.1735	−24970488	−24970487

(30)

The first ten rows give all endpoint signs in (28), hence the existence of a zero in every interval.

We next identify the root indices. The alternating series gives

$$J_1(r) > \frac{r}{2} - \frac{r^3}{16} \geq 0 \qquad (0 < r \leq \sqrt{8}).$$

The first inequality is strict even at $r = \sqrt{8}$, since the remaining alternating tail begins with a positive term and strictly decreasing absolute terms. Let a_1, a_2, a_3 be zeros in the three displayed J_1 intervals, in increasing order. Each interval has width less than $1 < 3 < \pi$, so Lemma A.1 makes its zero unique. There is no positive zero before a_1: such a zero would lie above $\sqrt{8} > 2$, and its distance to a_1 would be less than $3.83171 - 2 < 3 < \pi$.

If a further zero lay between a_1 and a_2, applying Lemma A.1 to the two subintervals would give $a_2 - a_1 > 2\pi$. This contradicts

$$a_2 - a_1 < 7.016 - 3.83170 = 3.18430 < 6 < 2\pi.$$

The same argument between a_2 and a_3 uses

$$a_3 - a_2 < 10.174 - 7.015 = 3.159 < 6 < 2\pi.$$

Thus $a_i = \kappa_i$ for $i = 1, 2, 3$. These are the only numerical uses of π: the elementary lower bound $\pi > 3$ suffices; no upper or finer lower bound is used.

For $\nu \geq 0$ the classical interlacing theorem is

$$j_{\nu,1} < j_{\nu+1,1} < j_{\nu,2} < j_{\nu+1,2} < \cdots \qquad \text{[14, Eq. 10.21.2]}.$$

For $\nu = 0$ it supplies exactly one zero of J_0 in $(0, \kappa_1)$ and exactly one in (κ_1, κ_2). The J_0 intervals in (28) lie in these respective gaps, so their zeros are q_1 and q_2, with uniqueness as asserted. □

Lemma A.3. *With $\alpha = q/k$, $\lambda = \alpha^2$, $c = -J_0(k)$, and $K = \alpha^2 J_1(q)^2/c^2$, one has*

$$.6276 < \alpha < .628, \qquad \frac{1}{4} < \lambda < \frac{2}{5}, \qquad .4027 < c < .4028, \qquad \frac{19}{5} < k < 4, \qquad K > .654.$$

Furthermore

$$B(\kappa_3) > .634, \qquad B(q_2) < .640, \qquad J_1(q_2)^2 < .116, \qquad k < \alpha\kappa_2 < q_2 < \alpha\kappa_3.$$

Proof. The integer-order integral representation [14, Eq. 10.9.2], differentiated for real x, gives

$$J_n(x) = \frac{1}{\pi}\int_0^\pi \cos(x\sin\vartheta - n\vartheta)\,d\vartheta, \qquad |J_n'(x)| \le 1 \quad (n \in \mathbb{Z}).$$

In particular, the last four rows of (30), the root intervals, and the mean-value theorem yield

$$J_1(q) > .51914 - .00001 > .5191, \qquad -.4028 < J_0(k) < -.4027,$$

$$|J_1(q_2)| < .34027 + .0001 < .3404, \qquad J_0(10.1735) < -.2497.$$

For example, the asserted two-sided bound at k follows already from

$$-.40275940 - .00001 < J_0(k) < -.40275939 + .00001.$$

The root intervals give

$$.6276 \cdot 3.83171 < 2.40482, \qquad 2.40483 < .628 \cdot 3.83170,$$

and hence the bounds on α. Since $q > 2$ and $k < 4$, one has $\lambda > 1/4$; and $.628^2 < 2/5$ proves its upper bound. Now

$$K > \frac{(.6276)^2(.5191)^2}{(.4028)^2} = \frac{66335796013041}{101404900000000} > .654,$$

$$J_1(q_2)^2 < (.3404)^2 = .11587216 < .116,$$

$$B(q_2) = q_2 J_1(q_2)^2 < 5.5202(.3404)^2 = .639637497632 < .640.$$

On the interval $(10.173, 10.174)$, uniqueness of the zero and the endpoint signs make J_1 positive before κ_3 and negative after it. Thus $J_0' = -J_1$ makes κ_3 the minimum of J_0 on the entire closed interval, and therefore

$$J_0(\kappa_3) \le J_0(10.1735) < -.2497.$$

It follows that

$$B(\kappa_3) = \kappa_3 J_0(\kappa_3)^2 > 10.173(.2497)^2 = .63428746557 > .634.$$

Finally the required order comparisons follow from the explicit rational chains

$$\begin{aligned} k &< 3.83171 < .6276 \cdot 7.015 = 4.402614 < \alpha\kappa_2, \\ \alpha\kappa_2 &< .628 \cdot 7.016 = 4.406048 < 5.5200 < q_2, \\ q_2 &< 5.5202 < .6276 \cdot 10.173 = 6.3845748 < \alpha\kappa_3. \end{aligned}$$

□

The same rational series bounds supply the few small-argument comparisons used earlier. At $r = 19/10$,

$$\frac{7}{25} < S_{0,3}(r) < J_0(r) < S_{0,4}(r) < \frac{283}{1000}, \qquad \frac{J_1(r)}{r} > \frac{S_{1,3}(r)}{r} > \frac{3}{10}.$$

Also

$$J_0(2) > S_{0,3}(2) = \frac{2}{9} > \frac{11}{50}, \qquad J_1(2) < S_{1,2}(2) = \frac{7}{12} < \frac{3}{5}.$$

All needed alternating tails are decreasing from their first omitted term. These estimates prove the asserted coarse bounds.

Appendix B. Zero extension, compactness and the disk spectrum

Lemma B.1. *Let $\Omega \subset \mathbb{R}^2$ be bounded and open. Zero extension identifies $H_0^m(\Omega)$ isometrically with a closed subspace of $H_0^m(Q)$, for $m = 1, 2$, whenever Q is a rectangle containing $\overline{\Omega}$ in its interior. There is a constant C_Q such that*

$$\|u\|_{H^2(\Omega)} \le C_Q \|\Delta u\|_{L^2(\Omega)}, \qquad u \in H_0^2(\Omega).$$

The embedding $H_0^2(\Omega) \hookrightarrow H_0^1(\Omega)$ is compact. Every element of $H_0^2(\Omega)$ has a continuous zero extension to $\mathbb{R}^2$, with bounded point evaluations. The buckling infimum is attained whenever Ω is nonempty.

Proof. Zero extension preserves the H^m norm on $C_c^\infty(\Omega)$; passing to the completion proves the first assertion and closedness. For smooth compactly supported functions, integration by parts gives

$$\|D^2 u\|_2^2 = \|\Delta u\|_2^2, \qquad \|\nabla u\|_2^2 = -\int u \Delta u.$$

Poincaré's inequality on Q first bounds $\|\nabla u\|_2$ by $C_Q \|\Delta u\|_2$, and then bounds $\|u\|_2$. Density gives the displayed coercive estimate. Rellich compactness on the fixed rectangle Q gives strong $H^1(Q)$ convergence of a subsequence from every bounded $H^2(Q)$ sequence, hence the asserted compactness. These are the usual embedding and compactness theorems, applied only on a rectangle; see [1, Chapters 4 and 6].

For completeness, point evaluation follows directly on the whole plane. With any fixed Plancherel normalization of the Fourier transform, Cauchy–Schwarz gives

$$\|\widehat{u}\|_{L^1(\mathbb{R}^2)} \le \left(\int_{\mathbb{R}^2} (1 + |\xi|^2)^{-2} \, d\xi \right)^{1/2} \|(1 + |\xi|^2)\widehat{u}\|_2 \le C \|u\|_{H^2(\mathbb{R}^2)}.$$

The first integral is finite. Fourier inversion therefore gives a continuous representative and $\|u\|_\infty \le C\|u\|_{H^2}$. This also proves uniform convergence of continuous representatives under H^2 convergence. Approximating by $C_c^\infty(\Omega)$ therefore shows that this representative is zero at every point of $\mathbb{R}^2 \setminus \Omega$, including its boundary. Finally, normalize the gradient energy of a buckling minimizing sequence to one. The coercive estimate, weak H^2 compactness and strong H^1 compactness give a minimizer with unit gradient energy. No extension theorem for the boundary of Ω is used. □

Lemma B.2. *On a disk $B_R(a)$, the first buckling eigenvalue is k^2/R^2, where $k = j_{1,1}$. Its eigenspace is one-dimensional and spanned by*

$$U_B(x) = J_0(k|x - a|/R) - J_0(k),$$

which is strictly positive in the disk.

Proof. Translation reduces to $a = 0$. A weak buckling eigenfunction satisfies $\Delta(\Delta + \Lambda)U = 0$. The distribution $(\Delta+\Lambda)U$ is harmonic, hence smooth; interior elliptic regularity for the second-order equation then makes U smooth in the disk. The relevant local regularity results can be found in [6, Sections 2.2 and 6.3].

Put $\rho = \sqrt{\Lambda} > 0$. The regular solution in angular mode $n \ge 0$ is a linear combination of r^n and $J_n(\rho r)$, with $r^0 = 1$ for $n = 0$. Indeed applying $\Delta + \Lambda$ to its regular harmonic component gives a particular solution proportional to r^n, and the remaining regular Helmholtz component is $J_n(\rho r)$. The two clamped traces at $r = R$ have a nonzero solution precisely when

$$\rho R J_n'(\rho R) - n J_n(\rho R) = -\rho R J_{n+1}(\rho R) = 0.$$

The trace assertion is the classical trace theorem on the smooth disk, applied to U and its first derivatives, which belong to $H_0^1(B_R)$ by the H_0^2 closure definition; see [7, Section 4.3]. Each

separated radial solution is smooth up to $r = R$, so these are ordinary endpoint values. The angular Fourier projections are bounded on $H^2(B_R)$: they are averages of rotations, which preserve the equivalent squared Sobolev norm $\|u\|_2^2+\|\nabla u\|_2^2+\|D^2u\|_2^2$. They preserve $H_0^2(B_R)$ by its closure definition, commute with the weak equation, and preserve the zero traces. Thus every nonzero Fourier component of an eigenfunction is subject to the displayed characteristic equation, and a nonzero eigenfunction has at least one such component. Interlacing of the positive zeros of consecutive Bessel orders gives $j_{1,1} < j_{2,1} < j_{3,1} < \cdots$; see [14, Section 10.21(i)]. All nonzero zeros are simple. Thus the smallest admissible ρR is the simple zero k in mode $n = 0$, which yields exactly the displayed function. The mode calculation also shows that no other angular component can belong to this eigenspace. The displayed U_B is an admissible clamped test function: near the boundary it satisfies $U_B = O((R-r)^2)$ and $\nabla U_B = O(R-r)$. Multiply it by a smooth cutoff that is zero for $R - r \le \varepsilon$ and one for $R - r \ge 2\varepsilon$. The error in each second derivative is uniformly bounded and supported in a collar of area $O(\varepsilon)$, so the cutoffs converge to U_B in H^2. Hence $U_B \in H_0^2(B_R)$. Its differential equation and integration by parts give its buckling quotient k^2/R^2, proving attainment of the stated smallest value. Finally $J_1 > 0$ on $(0, k)$ and $J_0' = -J_1$ imply its strict interior positivity. □

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